

Neel Gupta

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PROFESSIONAL SUMMARY

Mechanical Engineering student at UT Austin with hands-on experience building, testing, and troubleshooting mechanical hardware, embedded controls, and spacecraft systems.

EDUCATION

The University of Texas at Austin

Bachelor of Science in Mechanical Engineering | GPA: 3.76/4.00

Austin, TX

Expected May 2028

SKILLS

- **Design and Analysis:** SolidWorks, Siemens NX, AutoCAD, Autodesk Inventor, FEA, MATLAB, P&ID drawings, 3D printing
- **Engineering:** Mechanical design, hydraulic systems, testing and validation, root cause analysis, design for manufacturability (DFM), prototyping
- **Embedded and Controls:** Python, Arduino, Raspberry Pi, OpenCV, I2C, PWM, servo control, closed-loop control tuning, sensor integration and calibration

EXPERIENCE

NOV Inc.

Houston, TX

Mechanical Engineering Intern, Test Facility Modernization and Product Validation

May 2026 – Aug 2026

- Ran high-pressure qualification tests on hydraulic agitators and coiled tubing motors at multiple flow rates, measuring pressure and frequency to verify customer requirements.
- Mapped a coiled tubing dynamometer facility by tracing 200+ components across its high- and low-pressure circuits to recreate an undocumented system layout.
- Developed an Engineering Definition Package in AutoCAD with 8 drawings and documents, including P&IDs, general arrangement drawings, instrument lists, I/O lists, and a controls concept.
- Compared the facility with current standards and identified 15 upgrades for the modernization plan.
- Worked with mechanical, controls, and test engineers to troubleshoot hydraulic and sensor problems and update drawings to match the actual system.

University of Houston

Houston, TX

Research and Development Fellow

May 2025 – Aug 2025

- Built a pressure measurement system using 8 BMP581 sensors and a TCA I2C multiplexer to collect pressure readings across an airfoil.
- Tested the system in a wind tunnel to measure the airfoil's pressure distribution.
- Designed airfoil chambers with internal sensor channels in AutoCAD and built three 3D-printed versions, adjusting port tolerances until each sensor had an airtight seal.

PROJECTS

NASA L'SPACE Mission Concept Academy

Remote

Thermal Engineer, Lunar Rover Mission Design

May 2025 – Aug 2025

- Developed the thermal design for a lunar South Pole rover to keep the battery, avionics, and payload within a 0 to 40 °C operating range and a -20 to 50 °C survival range using MLI, thermal coatings, heat spreaders, and resistance heaters.
- Modeled radiator locations, heater zones, and thermal interfaces in Siemens NX, working with the mechanical team to fit the thermal hardware within the rover chassis and instrument deck.
- Worked with C&DH, power, science, and program management teams to coordinate thermal requirements and subsystem interfaces while keeping thermal power under 25 W average and 75 W peak.

Longhorn Baja SAE

Austin, TX

Ergonomics Subteam Member

Sep 2025 – May 2026

- Designed a competition steering wheel in SolidWorks to improve driver comfort and control while meeting manufacturing and vehicle packaging constraints.
- Validated the design with FEA and three 3D-printed prototypes, then worked with the machine shop to prepare the final design for machining.

AI Vision Tracking Camera System | SNDN Terrace LLC

Houston, TX

Embedded Systems Developer

May 2026 – Aug 2026

- Built a two-axis vision-tracking camera with Python and OpenCV on a Raspberry Pi 5, using a PCA9685 PWM driver over I2C to control the pan-and-tilt servos.
- Tuned a proportional controller that converts pixel tracking error into servo commands, using deadband and smoothing to reduce jitter and keep a moving target centered.
- Logged 1,600+ frames of tracking data at 28 fps and used response plots to tune controller gain and deadband, achieving median tracking errors of 7° in pan and 5° in tilt across a 160° pan range.